

PRIMARY ISSUE CATEGORY: STUCK PIPE

1. PROBLEM DEFINITION

Stuck pipe occurs when the drillstring cannot be moved axially (reciprocated) and/or rotated. Common mechanisms include differential sticking (pressure differential holds pipe against permeable formation), mechanical pack-off (cuttings bed or wellbore instability collapses around the BHA), and key-seating.

2. EARLY WARNING SIGNS

- Progressive increase in overpull (>1.4x baseline) across successive connections.
- Rising torque trend without a corresponding change in WOB/RPM.
- Erratic or reduced ROP with increased drag while reaming.
- Cuttings loading at surface increasing beyond expected hole-cleaning rate.

3. ROOT CAUSES (FIELD_ALPHA HISTORY)

- Inadequate hole cleaning in deviated/high-angle sections, allowing a cuttings bed to form.
- Reactive shale swelling on contact with water-based fluids, reducing annular clearance.
- Differential sticking against a permeable, underbalanced formation.
- Key seats formed at dogleg points from repeated string contact.

4. STEP-BY-STEP SOLUTION / RESPONSE PROCEDURE

Step 1: Stop and do not force it — Cease rotation/reciprocation attempts beyond safe overpull/torque limits to avoid parting the string.

Step 2: Establish circulation — Maintain or re-establish circulation at the maximum safe rate to improve hole cleaning and reduce differential sticking risk.

Step 3: Run a free-point survey — Use a free-point indicator (FPI) to identify the depth at which the string is genuinely free versus stuck.

Step 4: Spot a freeing pill — Spot a diesel/oil-based (or dedicated) freeing pill across the stuck interval to reduce friction and break differential sticking.

Step 5: Work the string — Apply controlled jarring/overpull cycles within string tensile limits; if unsuccessful, back off at the free point and prepare a fishing/sidetrack plan.

5. MATERIALS / PARAMETERS QUICK REFERENCE

Sticking Mechanism	Key Indicator	Primary Fix
Differential sticking	No torque increase, gradual overpull, stationary depth	Reduce MW/ECD, spot oil-based pill, work string
Mechanical pack-off	Torque AND overpull both rising, string fully immobile	Circulate, hi-vis sweep, jar, ream back carefully
Key seat	Overpull only on trip out, free when rotating	Ream through key seat, run key-seat wiper

6. PREVENTION CHECKLIST FOR OFFSET WELLS

- Increase hi-vis sweep frequency (e.g. every 150 m) through reactive shale / high-angle intervals.
- Set real-time torque/overpull trend alarms at 1.4x and 1.7x baseline.
- Schedule a preventive wiper trip if torque trend persists across 3+ connections.
- Keep pipe-freeing pill material pre-mixed and on standby when drilling known problem intervals.

7. FIELD CASE REFERENCES

Well	Depth	Outcome
W002	3160 m	Freed after 9 hr using FPI survey + 20 bbl diesel pill + jarring

This Issue Resolution Playbook is a synthetically generated reference document produced for the NWIS prototype, consolidating lessons across Field_Alpha and Field_Alpha-adjacent wells. Not associated with an actual drilling operation.